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| **Fiscal Year:** | FY 2006 | **Task Last Updated:** | FY 04/11/2006 |
| **PI Name:** | Rosenblatt, Charles S. Ph.D. | | |
| **Project Title:** | Simulated Microgravity Measurement Techniques for the Study of Dynamic Effects in Phospholipid Surfactants | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | FUNDAMENTAL PHYSICS | | |
| **Program/Discipline--Element/Subdiscipline:** | FUNDAMENTAL PHYSICS--Biological physics | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 00-HEDS-02 |
| **Start Date:** | 06/01/2001 | **End Date:** | 03/31/2006 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | 1 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA JPL |
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| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Taylor, Philip ( Case Western Reserve University ) | | |
| **Grant/Contract No.:** | NAG3-2910 | | |
| **Performance Goal No.:** | 1.24 | | |
| **Performance Goal Text:** | To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity | | |

We shall conduct a program of experiment and theory designed to provide rapid and reliable measurements of the dynamic properties of biologically important surfactants, such as pulmonary surfactants, under a wide variety of conditions. The essence of our technique rests on our ability to simulate a microgravity environment - indeed, we can simulate any environment between 0g and 1g, including those that vary with time - with a magnetic field so that we may perform resonance measurements on surfactant-coated liquid bridges. We will examine how the surface tension of an expanding or contracting air-water interface depends on the nature of the biological surfactants present and the rate at which the surface is expanding or contracting.
Using our magnetic levitation technique, we have undertaken an additional project, viz., the Rayleigh-Taylor and Saffman-Taylor instabilities. A less dense but paramagnetic fluid is pulled to the bottom of a tank by the magnetic field, and a more dense fluid rests on top. On turning off the field the fluid interface becomes unstable and the heavier fluid

**Task Description:**

falls to the bottom. The spatial period and growth rate of the surface perturbations will be measured. This work is being done in collaboration with Pierre Carlés of University of Paris VI. Because existing methods of measurement of dynamic surface tension have significant limitations, we have developed a sensitive technique that depends on the measurement of the resonance frequency of an oscillating liquid bridge, which requires a microgravity environment. Using a strong magnetic field gradient to simulate this environment, we will oscillate the field around this simulated microgravity set-point to induce motion in the bridge. As the resonance behavior of the bridge is exquisitely sensitive to a number of physical parameters, we can accurately determine the surface tension of the bridge. Moreover, by controlling the amplitude of the magnetic field oscillations, we can extract dynamic surface tension information, viz., how the surface tension of the surfactant-coated liquid changes with the rate of change of the liquid's surface area. These measurements can be performed at frequencies ranging from d.c. to 50 Hz, providing a dynamic range far greater than is possible with alternative techniques, and facilitating a physical understanding of the underlying dynamic surfactant behavior. Overall, our results will provide an enhanced understanding of the physical mechanisms involved in biological control of surface tension.

Tackling the two-fluid interface instability problem using magnetic levitation allows us to examine these instabilities in situ, without the need to use a moving tank and optical system. Moreover, we are able to tailor a specific temproal profile for the total body force (gravity + magnetic force). Thus, our objective is to improve our understanding and control of acceleration-driven fluid instabilities, especially in realms that are not now possible. Thus, with our technique, which facilitates very rapid changes in surface area, we wish to address several questions. What is the role of the minor lipids, such as PC, PG, and phosphatidylinositol in the process of DPPC adsorption in the absence of proteins? Specifically, how are the minor lipids "squeezed out" from the surface during the formation of the DPPC layer? On what time scales does this occur? [This question is particularly appropriate given the very large rate of change of area that we can generate in our system]. Other important questions include: How does overcompression affect the hysteresis behavior of the surface tension with area? It has been suggested that overcompression may result not only in single cycle hysteresis, but a general trend of increasing surface tension during subsequent cycling. We will therefore examine this issue not only in terms of the number of compression-expansion cycles, but also in terms of the frequency.

Fluid instabilities are found in many natural phenomena. To investigate the instabilities, current methods entail accelerating a tank containing two fluids, with the optical imaging equipment moving in tandem with the tank. This method is cumbersome, and does not facilitate investigation of a complex acceleration vs. time profile. Our magnetic levitation technique allows a much wider range of behavior, as the magnetic field can be programmed to exhibit a desired profile with time. For example, after partial development of the interfacial instability, the effective acceleration (i.e., the total body force) can be turned off -- or even reversed. We anticipate that our magnetic levitation method will provide important information about these instabilities that heretofore was not possible.

**Rationale for HRP Directed Research:**

**Research Impact/Earth Benefits:**

Our work on dynamic surface tension is related to medical issues such as respiritory distress syndrome in neonates, who often are unable to synthesize sufficient surfactant to facilitate pressure balance across the alveoli walls. We expect that our work will aid in understanding this problem, and provide information to create an appropriate cocktail of surfactants to alleviate this problem.
Our work on fluid instabilities allows us to examine and control the instabilityin regions of parameter space (e.g., large Capillary number) that is not feasible using extant methods. Since instabilities play important roles in many physical phenomena, including supernovas and even ink jet printing, our abilitiy to examine interface instabilities will facilitate a better understanding, and perhaps even improved technological capabilities, in a variety of related problems.

**Task Progress:**

We have tested the concept of magnetic levitation for interface instability studies, for which we examined the Rayleigh-Taylor instability in a Hele-Shaw cell]. Conditions for the Hele-Shaw regime, for which the velocity profile across the thickness of the cell must be Poiseuille-like at all times, can be met using sufficiently viscous fluids and a sufficiently thin cell of thickness h. Instabilities of interfaces between immiscible fluids have received scant attention compared to the case of miscible fluids because of the rapid destabilization of the interface. Previous methods that require physical movement of the cell are limited to slow processes, but fast destabilizations now can be studied using our magnetic levitation technique. To create a visual contrast between the two fluids we added a small amount of rubrene dye to chloroform, while highly paramagnetic manganese chloride tetrahydrate ($MnCl2{\cdot}4H2O$) was added to the weakly diamagnetic water in varying concentrations, rendering the aqueous solution paramagnetic. In the absence of an applied magnetic force the chloroform + dye sat at the bottom of the cell and the aqueous mixture at the top. With proper positioning of the cell, we turned on a downward magnetic force to create an artificially stable layering of the less dense but paramagnetic aqueous solution at the bottom of the cell and the more dense chloroform + dye at the top. Once the interface between the two layers stabilized, we switched off the magnet current, with the force decaying in a few tens of milliseconds. This subjected the fluids to ordinary gravity, at which point the fluid interface became unstable. We applied a code developed in our laboratory to locate the interfaces from the digitized images, which are shown by the dotted white lines. We then performed a four parameter fit of the interfaces and extracted the wavelength of the dominant perturbation and its time evolution, including its linear initial growth rate.
For immiscible fluids in a Hele-Shaw cell subjected to the Rayleigh-Taylor instability, the only extant model is that of Fernandez et al, which is applicable to a pair of fluids of the same viscosity — a rather difficult condition to fulfill with experimental constraints on their mutual miscibility, miscibility of a paramagnetic dopant, and densities. Motivated by the experimental results, our colleague Pierre Carlès led the development of a more general model. It was shown that this dispersion relationship can be simplified such that only two dimensionless numbers appear, the capillary number Ca and the normalized viscosity difference $\mu = (?1-?2)/(?1+?2)$, where $-1 = \mu = 1$. The growth rate is always real in the unstable case, i.e., the interface undergoes no temporal oscillations. If the two viscosities are equal (i.e., ?1 = ?2), we recover Fernandez's special case result. Our new result is applicable to fluids of arbitrary viscosities.

We found good agreement between experiment and theory. In particular, the comparison between the theoretical and experimental most unstable wavevectors k is excellent over the entire experimental range of Ca. This agreement confirms both the validity of the model and the precision of the experimental data acquisition. We stress that the quality of the experimental results is a direct consequence of the very precise control over the initial conditions obtained through magnetic levitation. It demonstrates the great advantage of this technique in the experimental study of gravitational interfacial instabilities.

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| | Another striking observation is the very small influence of reduced viscosity μ on the most unstable wavevector. Regardless of the viscosity difference, the two limiting curves are nearly identical. Nevertheless, one should not infer that the viscosity differences play no role: The dimensionless growth rate is normalized by the average of the two viscosities of the fluids, so that even if the dimensionless growth rate s seems little affected by the viscosity contrast, the dimensional growth rate is. The central result of this model is that, for the most unstable mode, a layering of immiscible fluids of different viscosities can be regarded almost as a layering of fluids of equal viscosities, having a value equal to the arithmetic mean of the two actual viscosities. Since our result shows that even pairs of fluids with high viscosity contrast can be considered a single viscosity system of appropriate value, to first approximation, the capillary number Ca remains the only significant experimental parameter. |
| **Bibliography Type:** | Description: (Last Updated: 12/04/2024) |
| **Articles in Peer-reviewed Journals** | Carles P, Huang Z, Carbone G, Rosenblatt C. "Rayleigh-Taylor instability for immiscible fluids of arbitrary viscosities: A magnetic levitation investigation and theoretical model." Phys. Rev. Lett. 2006 Mar;96(10):104501. PMID: 16605739 , Mar-2006 |